# Library walking access

## Phoenix, Arizona, USA

**5%** of residents can reach a library within a 15-minute walk (slope-aware travel time)

**1,446,730** people are beyond that 15-minute standard

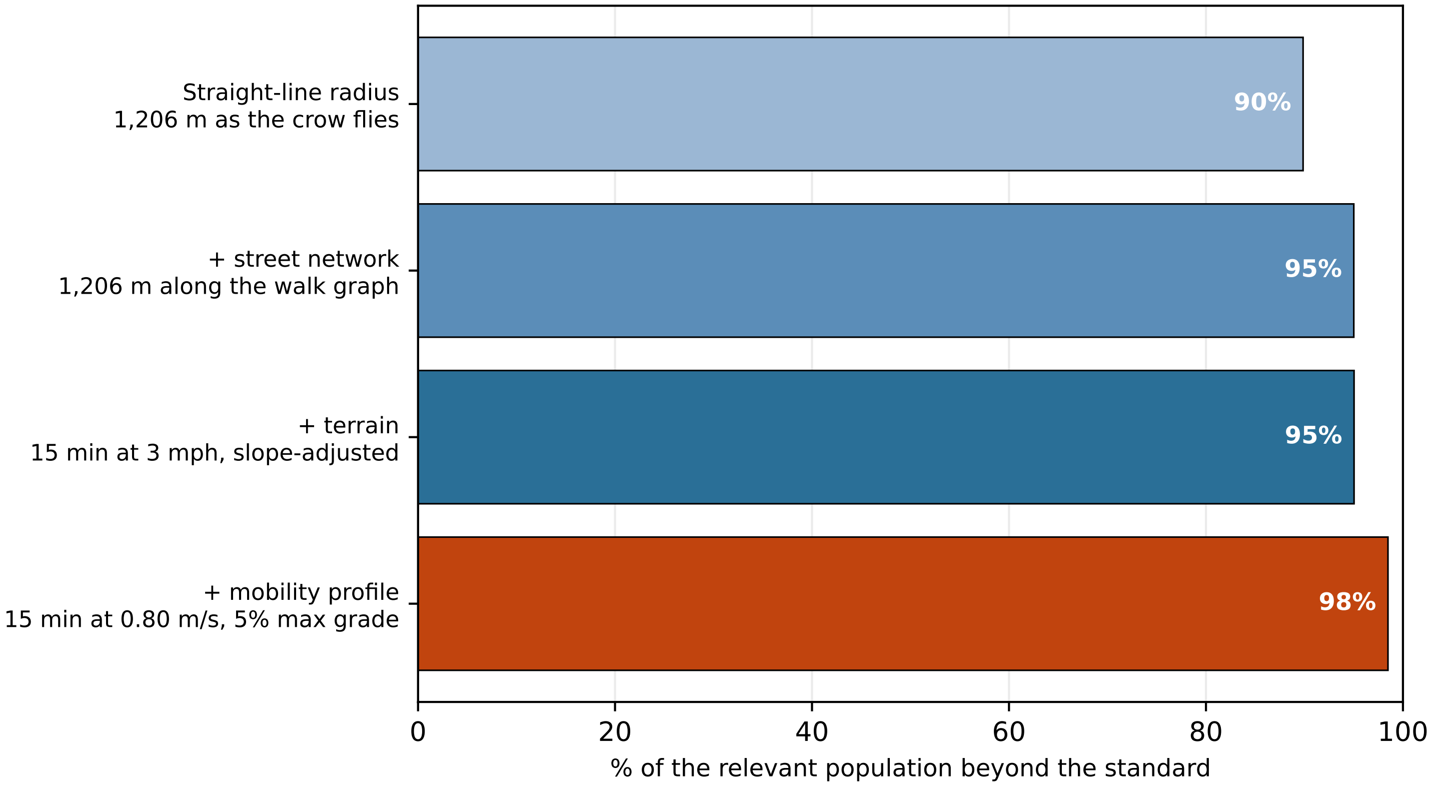
**98%** of residents with an ambulatory difficulty are beyond a 15-minute walk

**145 min** is the median walk faced by the worst-served of that group — before any slope penalty at all

Scope 17 library locations · 1,002 km² analysed · 1,522,291 residents
Terrain elevation 289–820 m · 1.9% of walk segments steeper than the 5% accessible-route grade
Standard 15-minute neighbourhood goal; 3 mph on the flat = 0.75 mi
Sources OpenStreetMap walk network · US Census ACS 5-year 2023 · USGS 3DEP elevation

# How you measure changes the answer

Each step is the same city and the same branches, measured more carefully




A straight-line radius — the way service areas are still often drawn — reports 90% of residents beyond a 15-minute walk. Routing along real streets and then charging for terrain puts it at 95%. Straight-line coverage understates the underserved population here by 5 points, or 78,795 people.

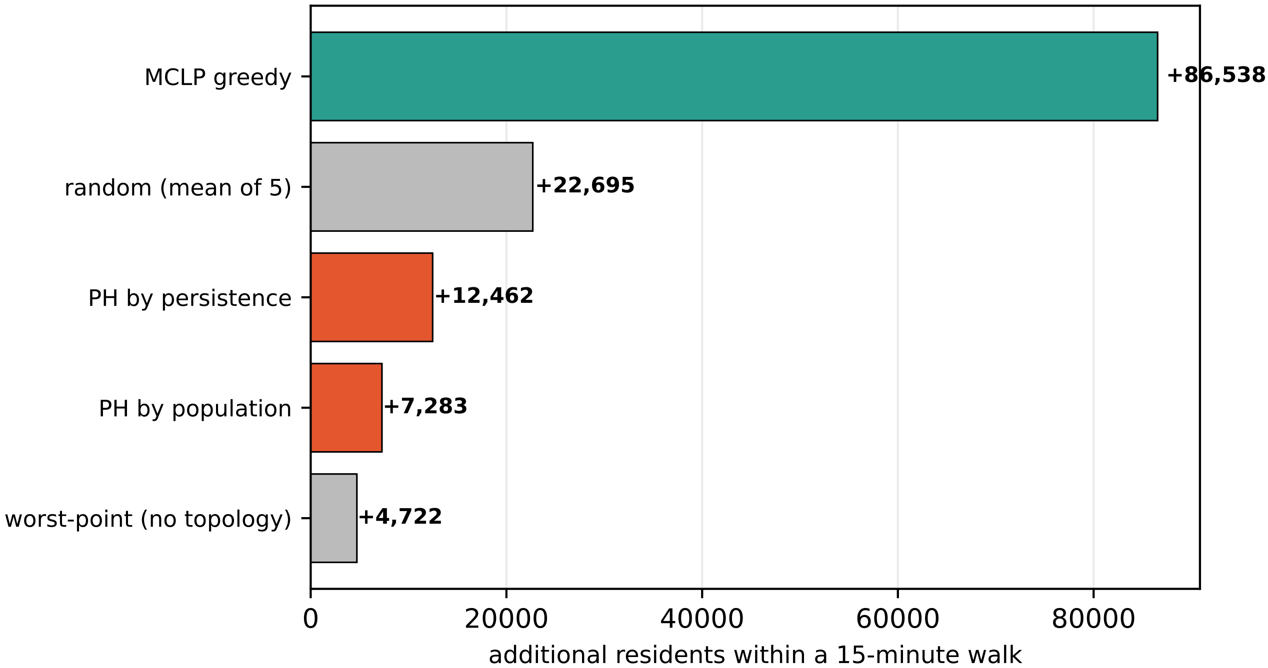
The first three bars hold the population fixed, so the differences are purely a matter of how access is measured. The fourth changes both the model and who it applies to: at 0.80 m/s with a 5% maximum grade, 98% of residents with an ambulatory difficulty are beyond the same standard.

## Which step matters most is a property of the city

In Phoenix the network step costs +5 points and terrain +0. Street networks impose a detour penalty everywhere — a grid forces Manhattan travel, roughly 1.3x straight-line — so the network step is large even in flat, regular cities. Terrain is what varies.

# Finding the gaps is not the same as filling them

Five strategies for siting 8 new branches, scored on residents brought within 15 minutes




This project began from a paper that uses topology — persistent homology — to find holes in civic coverage. It finds them. It is a poor guide to filling them.

Classical maximal-covering location (MCLP) adds 86,538 residents, raising coverage from 5% to 11%. Both topological strategies (12,462 and 7,283) land below five random draws (22,695). Topological holes are a diagnostic, not an optimiser.

## Why distance-driven strategies lose

Persistence marks the most remote point of a gap, and remoteness turns out to be a weak guide to how many people a branch would reach: across the 3,571 candidate sites the correlation between a site's travel time from existing branches and its marginal coverage gain is only -0.26 (rank correlation -0.25). The bulk relationship is weak — but the tail these strategies aim at is far worse than weak. A median candidate site gains 2,847 residents; the site the no-topology worst-point rule selects gains 91, or 3% of that. Random sampling draws from the middle of the distribution; targeting the extreme draws from its emptiest corner.

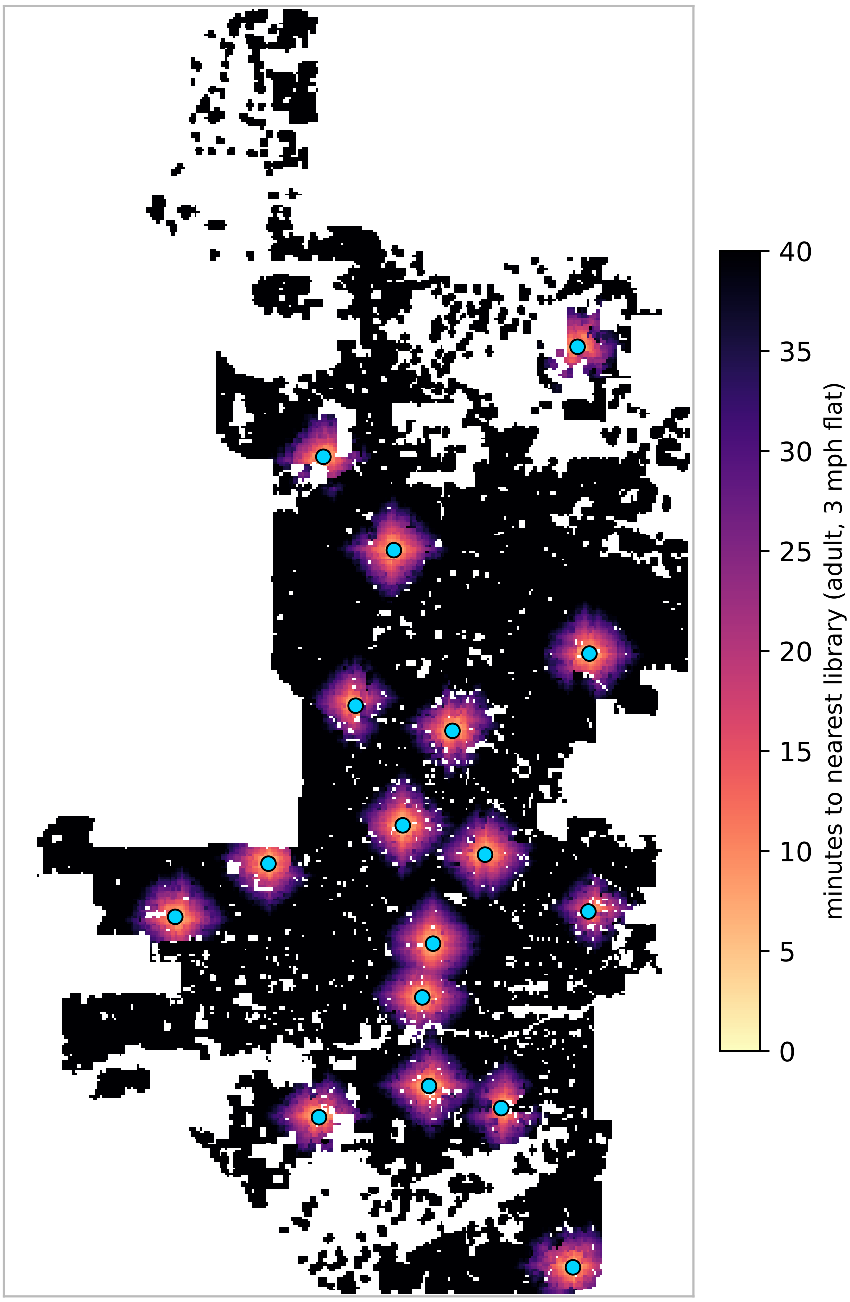
## How much the population ranking can help here

Very little in this city. Ranking gaps by population only discriminates if there is more than one gap worth ranking, and Phoenix's largest single pocket holds 98% of all underserved residents across 129 pockets — so the ranking returns the same pocket almost every time and the choice collapses back to picking its worst-served point.

Candidate sites are habitable cells on a 450 m grid (3,571 of them). Greedy MCLP is (1-1/e)-optimal for max-coverage, so an exact solve would only widen its margin.

# Where the walk is long

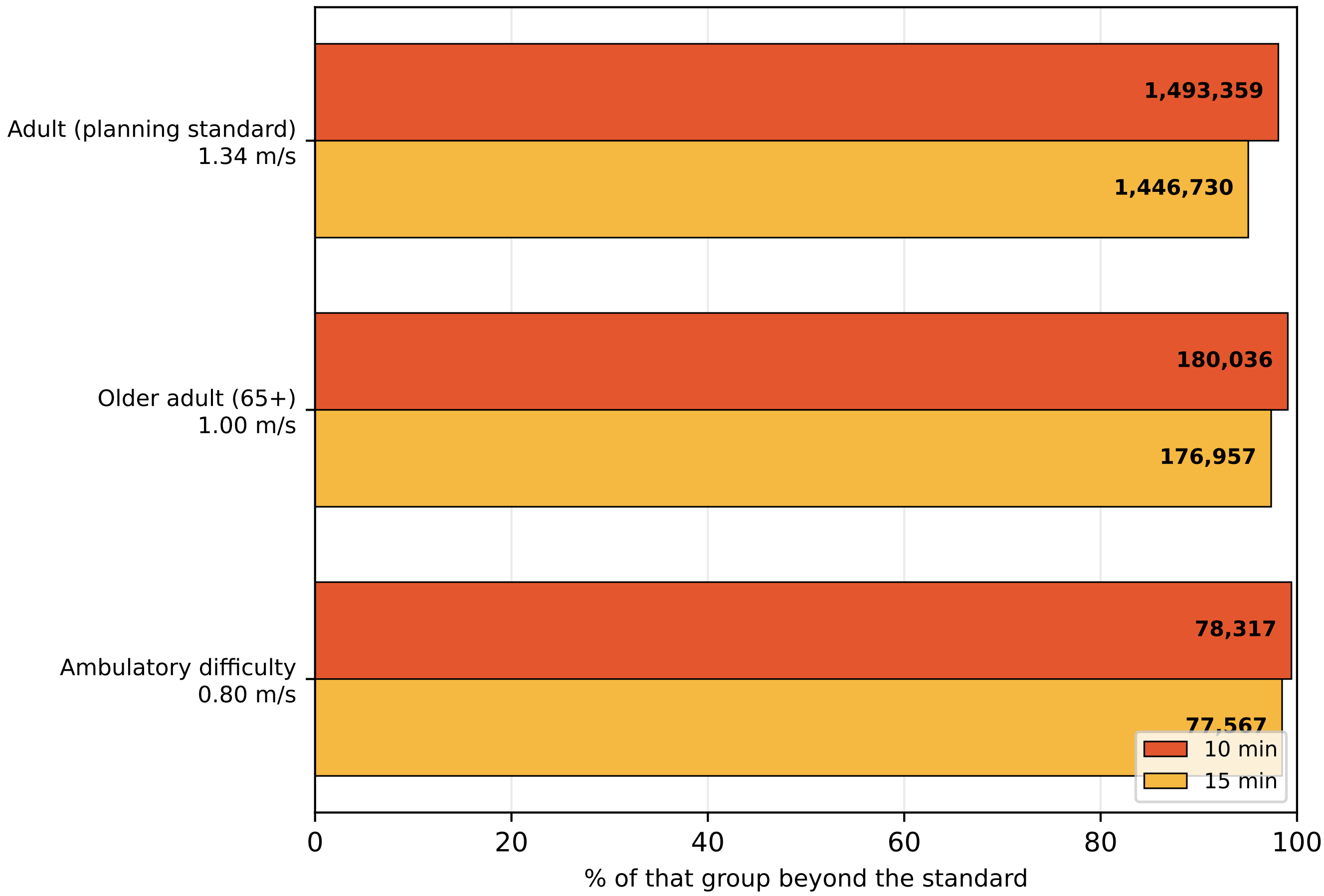
Travel time to the nearest library, on foot, accounting for slope




Median walk is 53 minutes. Blue dots are the 17 branches. Darker areas are further in time, not distance — a steep half-mile costs more than a flat one. Areas outside the analysed land mask (open water, and cells more than 250 m from any mapped pedestrian way) are blank.

# A time standard is not one distance

The same policy sentence describes a different city for each kind of walker

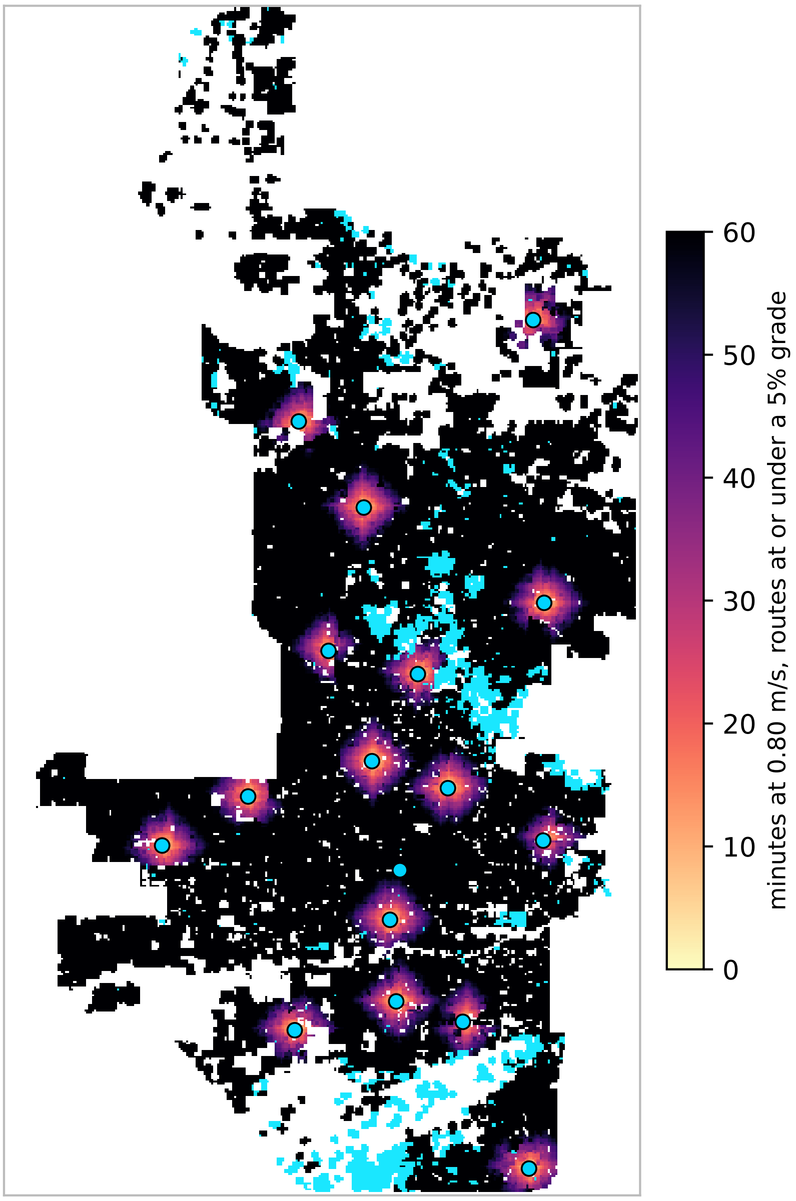



| profile | group | 10 min | 15 min | no route |
|---|---|---|---|---|
| Adult (planning standard) | 1,522,291 | 1,493,359 (98%) | 1,446,730 (95%) | 0 |
| Older adult (65+) | 181,742 | 180,036 (99%) | 176,957 (97%) | 0 |
| Ambulatory difficulty | 78,765 | 78,317 (99%) | 77,567 (98%) | 1,063 |

Each row is measured against its own population, not a share of the city. Walking speeds are planning defaults: 3 mph adult, 1.00 m/s for 65+, 0.80 m/s with an ambulatory difficulty.

# Slope, and who it excludes

Measured against the 1:20 (5%) grade an accessible route is designed to



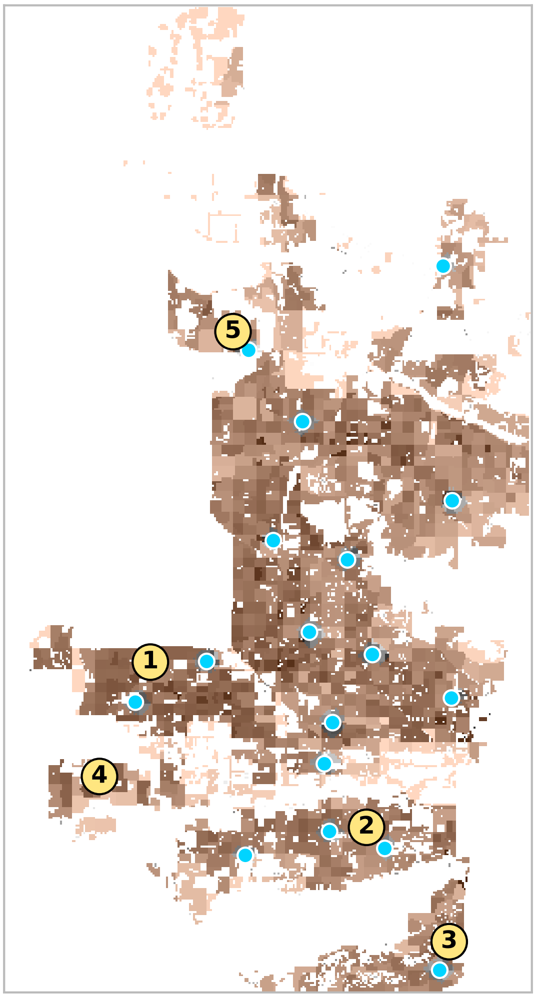

Cyan marks land with no route to any library that stays within a 5% grade: 1,063 residents with an ambulatory difficulty, 1% of that group. 1.9% of walk segments are steeper than that.

That count is sensitive to the threshold — it ranges 79–1,628 across cutoffs from 4% to 15% — so it should not be read as a precise figure. Two things underneath it are not sensitive.

These are remote places before grade is considered at all: with no slope penalty whatsoever the same cohort still faces a median 145-minute walk, and 87% over an hour. And the burden falls almost entirely on this profile — grade moves the adult figure about two points, against 0 points here.

# Underserved pockets and best sites

Contiguous residential areas beyond a 15-minute walk, ranked by residents affected




| # | residents | car-free HH | area | worst walk | site gains |
|---|---|---|---|---|---|
| 1 | 1,083,825 | 30,195 | 481.3 km² | 198 min | 12,582 |
| 2 | 146,326 | 2,616 | 72.0 km² | 148 min | 7,226 |
| 3 | 61,004 | 730 | 29.4 km² | 122 min | 5,463 |
| 4 | 41,176 | 44 | 18.6 km² | 213 min | 6,905 |
| 5 | 27,700 | 18 | 17.1 km² | 128 min | 4,368 |
| 6 | 19,728 | 245 | 15.1 km² | 155 min | 4,074 |

Numbered markers are the best available branch site inside each pocket — the location that brings the most residents within 15 minutes — chosen from habitable land only, so parks, port terminals and airfields are never proposed.

**These are not real parcels.**

A marker means “somewhere around here would help most”, resolved to a 150 m grid. It carries no assessment of zoning, ownership, lot size, acquisition cost, or whether anything is buildable on the site. Treat each one as a search area for a siting study, not a candidate address.

# Method and known limits

What this models, and what it does not

Walk network from OpenStreetMap, retaining all connected components — a city's pedestrian network is generally not one component, and the default of keeping only the largest silently deletes whole districts. Travel time uses Tobler's hiking function renormalised to each profile's flat speed, over USGS 3DEP elevation at 15 m. Population is ACS 2023 block-group data allocated onto habitable land; poverty, vehicle access and ambulatory difficulty are tract-level and coarser.

## The 5% grade is a threshold, and thresholds are a modelling choice

The mobility profile treats segments steeper than 5% as impassable. Real mobility is not binary, and this was chosen for tractability and legibility rather than realism. Two things bound the consequences. The count is sensitive to where the line is drawn — it ranges 79–1,628 across cutoffs from 4% to 15%, so it should never be quoted alone. But under a graded penalty instead of a cutoff, nobody is stranded and the same cohort still faces a median 145-minute walk with no slope penalty applied at all. The exclusion is structural remoteness compounded by terrain, not an artefact of the threshold.

## Not modelled

Sidewalk presence and quality, curb ramps, crossing delay, cross slope (capped at 2.1% under PROWAG and a common real-world failure), transit, opening hours and branch capacity. Grade is taken from street centrelines, not sidewalks. Every one of these omissions pushes the same way: the mobility figures here are optimistic.

Walking speeds are planning defaults, not locally observed. Travel time is one-way — a downhill outbound trip is uphill on the return. Facility sets are branch locations only.

## Sources

OpenStreetMap (walk network, land use) · US Census ACS 5-year 2023 and TIGER (population, demographics, water) · USGS 3DEP (elevation) · municipal open data (branch locations).

Grade thresholds follow the 2010 ADA Standards and the US Access Board's PROWAG: an accessible route is limited to 1:20 (5%); 1:12 (8.3%) applies to ramps and curb ramps, not to walking a block. PROWAG permits a route to match the adjacent street grade where that exceeds 5%, so a steep sidewalk may be compliant — 5% is the grade an accessible route is designed to, which is why this report says “within a 5% grade” and not “ADA-compliant”.